gf

2023-03-30

```
## tibble [53,940 x 10] (S3: tbl_df/tbl/data.frame)
##  $ carat  : num [1:53940] 0.23 0.21 0.23 0.29 0.31 0.24 0.24 0.26 0.22 0.23 ...
##  $ cut    : Ord.factor w/ 5 levels "Fair"<"Good"<..: 5 4 2 4 2 3 3 3 1 3 ...
##  $ color  : Ord.factor w/ 7 levels "D"<"E"<"F"<"G"<..: 2 2 2 6 7 7 6 5 2 5 ...
##  $ clarity: Ord.factor w/ 8 levels "I1"<"SI2"<"SI1"<..: 2 3 5 4 2 6 7 3 4 5 ...
##  $ depth  : num [1:53940] 61.5 59.8 56.9 62.4 63.3 62.8 62.3 61.9 65.1 59.4 ...
##  $ table  : num [1:53940] 55 61 65 58 58 57 57 55 61 61 ...
##  $ price  : int [1:53940] 326 326 327 334 335 336 336 337 337 338 ...
##  $ x      : num [1:53940] 3.95 3.89 4.05 4.2 4.34 3.94 3.95 4.07 3.87 4 ...
##  $ y      : num [1:53940] 3.98 3.84 4.07 4.23 4.35 3.96 3.98 4.11 3.78 4.05 ...
##  $ z      : num [1:53940] 2.43 2.31 2.31 2.63 2.75 2.48 2.47 2.53 2.49 2.39 ...
```

Question 1

Assumptions

1. The data is made up of basic random values drawn from the population
2. First, we test the hypothesis of normality of the dependent variable therefore the price of diamonds

```
##
##  Shapiro-Wilk normality test
##
## data:  diamonds$price[1:5000]
## W = 0.66175, p-value < 2.2e-16
```

Since the p-value $<$ 2.2e-16, implies that data distribution don't follows the normal distribution.

3. Next, we will test the hypothesis of equality of variances between the groups.

```
##
##  Bartlett test of homogeneity of variances
##
## data:  diamonds$price by diamonds$cut
## Bartlett's K-squared = 406.7, df = 4, p-value < 2.2e-16
```

Since the p-value$<$ 2.2e-16, implies that the variances don't to be equal.

ANOVA

Let m1, m2, m3, m4 and m5 respectively be the mean prices of diamonds which group are "Ideal", "Premium", "Good", "Very Good", and "Fair".

H0: m1=m2=m3=m4=m5 Ha: At least one mean different

```
##
##  One-way analysis of means (not assuming equal variances)
##
## data:  price and cut
## F = 166.04, num df = 4.0, denom df = 9398.6, p-value < 2.2e-16
```

Conclusion With an alpha of 0.05, and a p-value< 2.2e-16, we reject the null and stat that we have sufficient evidence to show that there was a effect of cut on diamond prices.

Question 2

The hypotheses are:

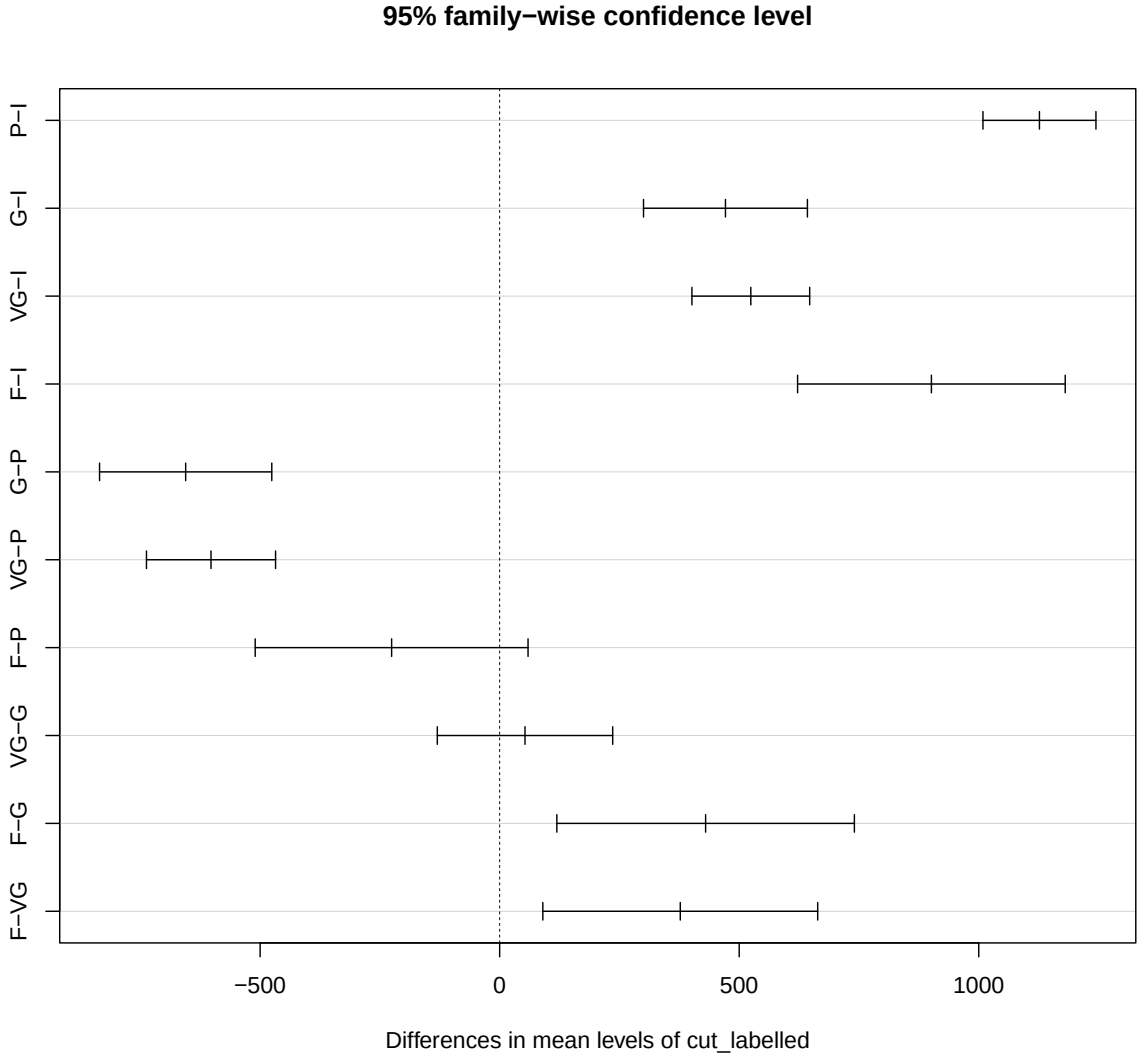
- H0: m1=m2=m3=m4=m5
- Ha: At least one mean different

```
##
##  One-way analysis of means
##
## data:  price and cut
## F = 175.69, num df = 4, denom df = 53935, p-value < 2.2e-16
```

There is no obvious difference in p-value but the F-statistic is different as well as "num df" and "denom df".

Question 3

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = price ~ cut_labelled, data = diamonds)
##
## $cut_labelled
##            diff       lwr        upr     p adj
## P-I  1126.71573 1008.8088 1244.62267 0.0000000
## G-I   471.32248  300.2823  642.36268 0.0000000
## VG-I  524.21792  401.3312  647.10467 0.0000000
## F-I   901.21579  621.8602 1180.57139 0.0000000
## G-P  -655.39325 -835.1353 -475.65120 0.0000000
## VG-P -602.49781 -737.2331 -467.76249 0.0000000
## F-P  -225.49994 -510.2665   59.26664 0.1950425
## VG-G   52.89544 -130.1519  235.94273 0.9341158
## F-G   429.89331  119.3378  740.44880 0.0014980
## F-VG  376.99787   90.1336  663.86215 0.0031094
```



Interpretations

Question 4

The hypotheses are:

- H0: m3=m4
- Ha: m3!=m4

```
##
##  One-way analysis of means
##
## data:  price and cut
## F = 0.65381, num df = 1, denom df = 16986, p-value = 0.4188
```

Conclusion With an alpha of 0.05, and a p-value of 0.4188, we fail to reject the null and stat that we have insufficient evidence to show that there was a cut effect of on diamond prices.

Question 5

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = price ~ cut, data = df)
##
## $cut
##                     diff       lwr      upr     p adj
## Very Good-Good 52.89544 -75.32923 181.1201 0.4187652
```

